#25 Autonomous AI Web Crawler

Research / Autonomous Web Intelligence

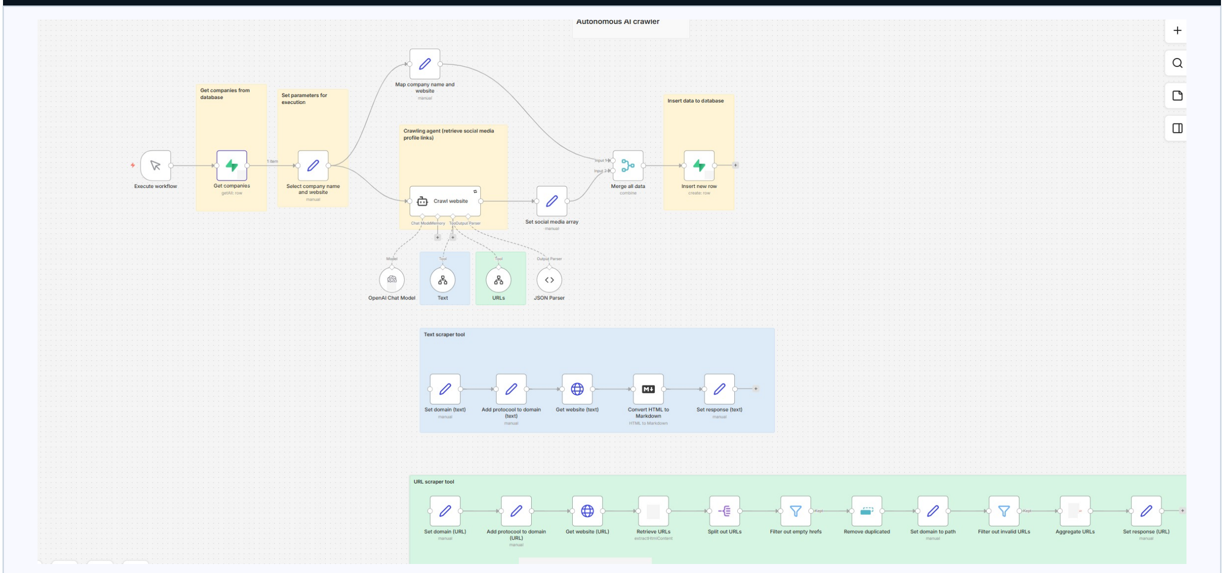
Self-directing agent maps company social footprints using sub-workflow tools

BUSINESS PROBLEM

Sales teams need competitor digital intelligence but manual research takes hours per company. This crawler autonomously visits each company homepage, extracts all external links, identifies social platforms, and builds a complete social footprint record — processing 100 companies overnight.

n8n CANVAS — actual workflow screenshot

n8n WORKFLOW EDITOR Autonomous AI crawler.json



Execute Workflow → Get Companies (Supabase) → Select Company → Crawl Website Agent → Set Social Array → Merge All Data → Insert New Row (Supabase) | Sub-tools: Text Scraper + URL Scraper

INPUT / OUTPUT — exact n8n JSON format

INPUT — Execute workflow trigger — company list batch from Supabase


```
{
  "batch_size":   50,
  "filter":       "target_account = true",
  "crawl_depth":  2,
  "timeout_sec":  30,
  "platforms":    ["linkedin","twitter","github","youtube"]
}
```


OUTPUT — Supabase upsert — enriched company record with social footprint


```
{
  "json": {
    "company_id":     "comp-0441",
    "company_name":   "TechScale Solutions",
    "website":        "techscale.io",
    "linkedin_url":   "linkedin.com/company/techscale-solutions",
    "linkedin_followers": 4820,
    "twitter_handle": "@TechScaleHQ",
    "posts_per_week": 8,
    "primary_topics": ["DevOps","Kubernetes","Platform
Engineering"],
    "tone":           "technical-practical",
    "crawled_at":     "2024-06-15T02:14:33Z"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ **Execute Workflow — start crawl batch**	Manual or cron trigger. Parameters: filter criteria, crawl depth, social platforms to detect.
2	■ **Get Companies — Supabase fetch**	SELECT from companies WHERE target_account=true. Returns: {company_id, name, website_url, industry}.
3	■ **Select Company — extract name and URL**	Isolates company_name and website_url as clean variables for the crawling agent tool calls.
4	■ **Crawl Website Agent — AI orchestration**	GPT-4 agent iteratively calls Text tool and URLs tool: visit homepage, extract links, identify social profiles.
5	■ **Text Scraper Tool (sub-workflow)**	Set domain → Add protocol → GET website text → Convert HTML to Markdown → Return structured summary.
6	■ **URL Scraper Tool (sub-workflow)**	Set domain → GET website → Extract URLs → Split Out → Filter empty → Remove duplicates → Filter invalid → Aggregate → Return URL list.
7	■ **OpenAI GPT-4 — social identification**	Identifies which extracted URLs are social platform profiles vs ads, partner links, or CDN assets.
8	■ **JSON Parser — structured output**	Forces schema: {linkedin_url, twitter_handle, github_url, followers, posts_per_week, primary_topics, tone}.
9	■ **Merge All Data — combine sources**	Joins original Supabase company record with all crawled social intelligence into single enriched object.
10	■ **Insert New Row — upsert to Supabase**	Upserts enriched record: social URLs, follower counts, posting frequency, topics, last_crawled_at.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com	OpenAI API
Supabase URL + Service Role Key	app.supabase.com → Settings → API → service_role key	Supabase API

SAMPLE RUN — what this workflow processed and produced

Scenario	A B2B SaaS company maintains 200 target accounts. Sales team needed social intelligence for personalised outreach but manual research took 15-20 minutes per company.
Input received	Sunday 02:00 cron. Supabase returned 50 companies marked target_account=true. Processing started sequentially.
What the workflow found	TechScale Solutions homepage crawled in 3.2s. URL scraper extracted 847 links. After filtering: 6 social platform URLs identified. LinkedIn: 4,820 followers, 8 posts/week, topics: DevOps, Kubernetes.
Output produced	50 records enriched in 18 minutes 42 seconds. Sales personalised outreach referencing recent LinkedIn posts increased reply rate from 4% to 11%.

JSON: Autonomous AI crawler.json	**GitHub: trm-autonomous-web-crawler**	apify.com/store